10.4.3 Nested if

An if statement which contains another if statement is called nested if. The nested can have one of the following three forms.

1. If nested inside if part
2. If nested inside else part
3. If nested inside both if part and else part

The syntax of the nested if:

If nested inside if part

```
if (expression-1)
{
    if (expression-2)
    {
     True_Part_Statements;
    }
    else
    {
     False_Part_Statements;
    }
}
else
     body of else part;
```

If nested inside else part

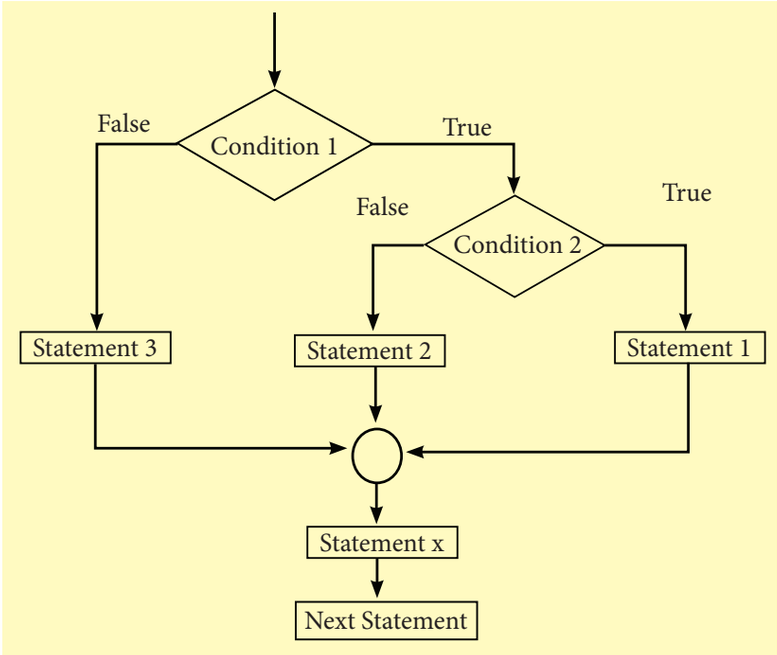
```
if (expression-1)
{
     body of true part;
}
else
{
     if (expression-2)
     {
      True_Part_Statements;
     }
     else
     {
      False_Part_Statements;
     }
}
```

If nested inside both if part and else part

```
if (expression)
{
     if (expression)
     {
      True_Part_Statements;
     }
     else
     {
      False_Part_Statements;
     }
}
else
{
     if (expression)
     {
      True_Part_Statements;
     }
     else
     {
      False_Part_Statements;
     }
}
```

In the first syntax of the nested if mentioned above the expression-1 is evaluated and the expression result is false then control passes to statement-m. Otherwise, expression-2 is evaluated,if the condition is true, then Nested-True-block is executed, next statement-n is also executed. Otherwise Nested-False-Block, statement-n and statement-m are executed.

The working procedure of the above said if..else structures are given as flowchart below:



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